CriterionLens: Evidence-Grounded Criterion-Level Clinical Trial Matching with Small Language Models

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***Abstract*—Clinical-trial eligibility matching requires a separate, evidence-grounded judgment for each criterion, including an explicit way to abstain when a note is incomplete. CriterionLens fine-tunes compact instruction models to return a structured label and numbered supporting sentences, and pairs the screening pipeline with explicit KV-cache, tiled-attention, and greedy speculative-decoding implementations. On the fixed held-out split, the fine-tuned 1.5B model obtains accuracy 0.7621 and fixed-label macro-F1 0.4762 (bootstrap 95% intervals [0.704, 0.816] and [0.420, 0.529]). On a Tesla T4, the KV cache speeds up 1.5B generation by 9.8$imes$ at a 1024-token prompt, tiled attention matches the reference within 10^{-3} in float16 while completing an 8192-token case on which the naive reference runs out of memory, and speculative decoding is exact but, with a 0.5B draft, slower than target-only decoding (0.60–0.94$imes$). These results characterize a small research prototype on a limited annotated dataset; they do not establish clinical utility. The official application is browser-accessible and locally hosted, as permitted by the assignment.**

***Index Terms*—clinical trial matching, evidence extraction, language models, parameter-efficient fine-tuning, inference**

I. Introduction

Clinical-trial screening compares a patient's clinical history with many protocol eligibility criteria. A trial-level decision can hide which individual requirement was supported, contradicted, or left unknown. Prior work has therefore framed matching as criterion-level classification and explanation [1], [2]. CriterionLens follows that framing but targets a small, locally runnable decoder and returns sentence identifiers rather than an unconstrained rationale.

The system has four contributions. First, it fine-tunes Qwen2.5 instruction models with completion-only LoRA updates and evaluates them on a patient-disjoint split. Second, it extracts numbered evidence, validates output schemas, and treats missing information and unsupported decisive labels conservatively. Third, it implements and measures explicit KV-cached generation, tiled online-softmax attention, and greedy speculative decoding. Fourth, it packages these components in a local Gradio application that is explicitly a research aid, not a medical device. Figure 1 summarizes how a criterion-level prediction is checked and routed for human review.

II. Problem formulation

Given a numbered patient note x and one inclusion or exclusion criterion c, the model predicts an allowed criterion-type-specific label y and a set of supporting sentence IDs E. The label includes a not-enough-information option; absent evidence is not interpreted as proof of eligibility or ineligibility. A later aggregation rule produces only a preliminary trial-level summary and routes unresolved cases to manual review.

III. Related Work

A. Patient-to-trial matching

TrialGPT structures matching around criterion-level judgments and explanations [1]. The SIGIR clinical-trial collection provides a reproducible benchmark for patient-to-trial matching [2]. These resources motivate explicit criterion-level output, but benchmark performance alone cannot establish safe enrollment decisions. CriterionLens therefore exposes evidence and abstention and makes no clinical-use claim.

B. Transformers and parameter-efficient adaptation

Self-attention provides parallel sequence modeling but its full attention matrix scales quadratically with sequence length [3]. Qwen2.5 is an open-weight instruction-model family [4]; its rotary positions and grouped-query attention inform the decoding implementation [5], [6]. LoRA learns low-rank updates while freezing base weights [7]. QLoRA is a memory-saving alternative, but was not required for the selected training runs [8].

C. Efficient inference

The KV cache avoids recomputing prior keys and values during autoregressive generation; grouped-query attention reduces its size. Online softmax and FlashAttention motivate tiled attention that avoids materializing the full score matrix [9], [10]. PagedAttention addresses serving-time cache management at a different systems level [11]. Speculative decoding uses a draft model and target verification, with realized speed depending on acceptance and hardware [12]. The implementations here are measured independently; theoretical efficiency is not treated as an empirical speedup.

IV. Data and Model Selection

A. Dataset and audit

The experiments use the public TrialGPT criterion-annotation dataset [1]. The audit retained 1014 annotations from 53 patients and 103 trials after dropping empty criterion text. GPT-4-generated explanations and labels were excluded

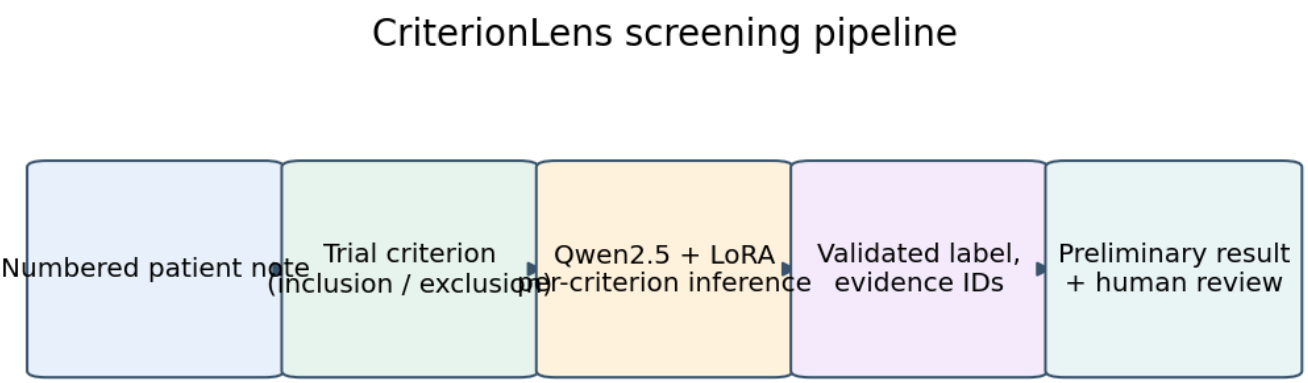


Fig. 1. CriterionLens screening pipeline from numbered note and criterion to a validated preliminary result.

TABLE I
PATIENT-DISJOINT DATASET PARTITION SIZES.

Split	Patients	Annotations	Inclusion	Exclusion
TRAIN	37	602	215	387
VAL	8	206	66	140
TEST	8	206	75	131

TABLE II
LORA TRAINING CONFIGURATION READ FROM THE VERSIONED YAML FILES.

Model	Max tokens	Epochs	Batch x accum.	Rank/alpha	Learning rate	Seed
1.5B	640	3	4 x 4	16/32	2.0e-04	42
0.5B	640	3	4 x 4	16/32	2.0e-04	42

at load time; training targets use only expert eligibility labels and expert evidence sentence IDs. The fixed split is patient-disjoint, with seed 42 and counts shown in Table I.

The split is frozen by a manifest containing the dataset revision and patient assignments. All prompt, training, and model comparisons use the same split; the held-out test partition is accessed only by the tagged final evaluation.

B. Models and supervised fine-tuning

Qwen2.5-1.5B-Instruct is the primary target and Qwen2.5-0.5B-Instruct is the size comparison and same-tokenizer draft [4]. LoRA updates query, key, value, output, and MLP projections while the base weights remain frozen [7]. The training target is a compact JSON object with evidence IDs followed by a criterion-type-specific label. The patient note is placed before the criterion; explanations are generated from cited text by a deterministic template, not learned from GPT-4 columns.

Training histories record optimization loss and validation loss by epoch. They do not contain generation-based validation F1 for every checkpoint. Therefore, the report compares the completed final adapters on the already-run fixed validation evaluations and does not claim an epoch was selected by generation F1 (D-006). The versioned training settings are summarized in Table II. Figure 2 shows the prompt/target construction and the response-only loss mask.

V. CRITERION-LEVEL INFERENCE AND DECODING

A. Structured prediction and evidence

The instruction prompt requests one JSON object. The parser validates the label against the criterion type, removes and records out-of-range sentence IDs, and invokes constrained label scoring if generation cannot be parsed. A decisive prediction without evidence is marked unsupported. A grounded template explains the output using only cited sentences; the aggregation rule treats unsupported or unknown criteria as requiring review rather than as a pass.

B. KV-cached autoregressive decoding

The reference decoder recomputes the full prefix. The cached decoder prefills a Hugging Face dynamic cache once, then forwards one token per step with the absolute position and an expanded attention mask. With L layers, H_{kv} key/value heads, head dimension d_h, sequence length S, batch B, and b bytes per element, the cache allocation is

$$M_{KV} = 2LBH_{kv}Sd_hb. \tag{1}$$

The factor two accounts for keys and values. Tests check token equality on tiny models, cache growth, byte counts, and a negative position-reset case. Timing and allocation are measured separately from correctness. Figure 3 shows that the cache is extended with each token and reused by the next decode step.

C. Tiled online-softmax attention

The tiled implementation traverses query and key blocks and maintains a running row maximum m, normalizer ℓ, and weighted-value accumulator a. When a new key tile

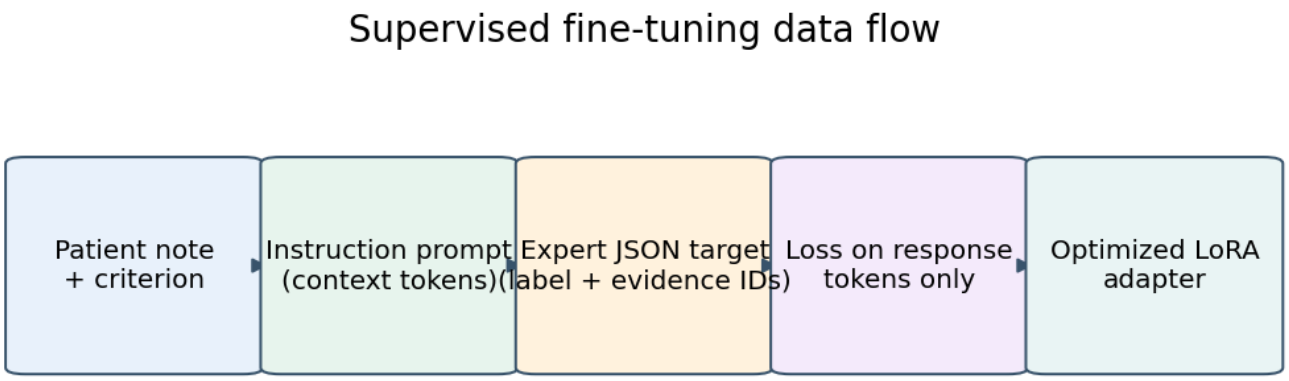


Fig. 2. Fine-tuning flow using expert labels and evidence IDs, with loss applied to response tokens only.

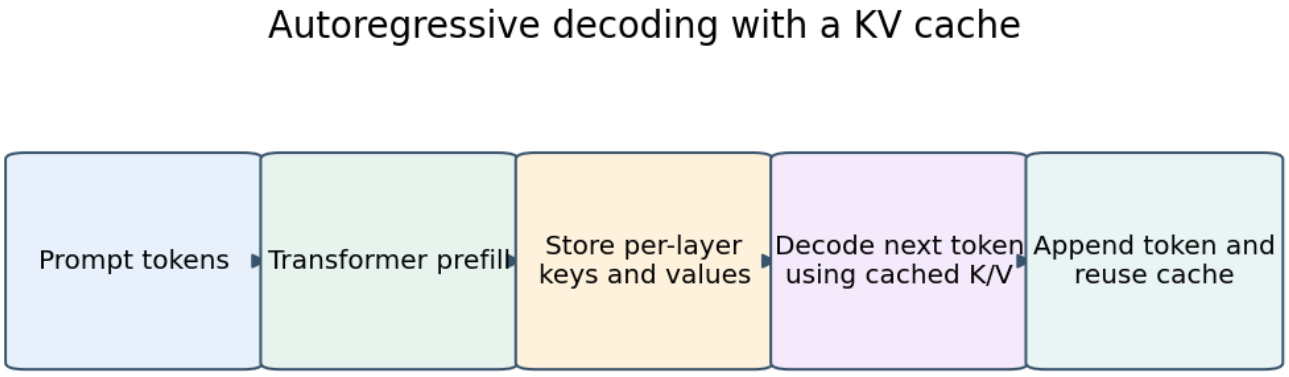


Fig. 3. KV-cached autoregressive decoding: the prefix is prefetched once and each generated token extends the per-layer key/value cache.

changes the maximum from m to m', prior accumulators are rescaled by $\alpha = \exp(m - m')$, and the tile contributes its own normalized weights. The final output is a/ℓ. This avoids storing the full score matrix while keeping accumulation in higher precision. Numerical error is compared with a double-precision reference; fused SDPA methods are performance baselines, not the implementation under test. Figure 4 illustrates the online rescaling that permits blockwise accumulation without materializing the full score matrix.

D. Greedy speculative decoding

The smaller model proposes up to k tokens, then the target verifies the proposed block. Matching target argmax tokens are accepted; at the first mismatch the target token is emitted and the draft cache is cropped to the accepted prefix. The method is greedy and is checked for exact output equality against target KV-cache decoding. Acceptance rate and timing are reported independently because matching tokens need not yield a speedup on every hardware configuration. Figure 5 depicts the accepted-prefix and target-correction steps for greedy decoding.

VI. Experimental Setup and Metrics

The training and benchmark kernels requested two NVIDIA Tesla T4 GPUs on Kaggle. Each benchmark process measures one model at a time; the two-GPU allocation is not described as data-parallel inference. The laptop environment was used for unit tests and application work. The exact software versions, model identifiers, dtype, device, timing repetitions, and input source are retained in the result JSON files.

Classification reports accuracy, observed-label macro-F1, fixed-label macro-F1, per-class scores, and confusion matrices.

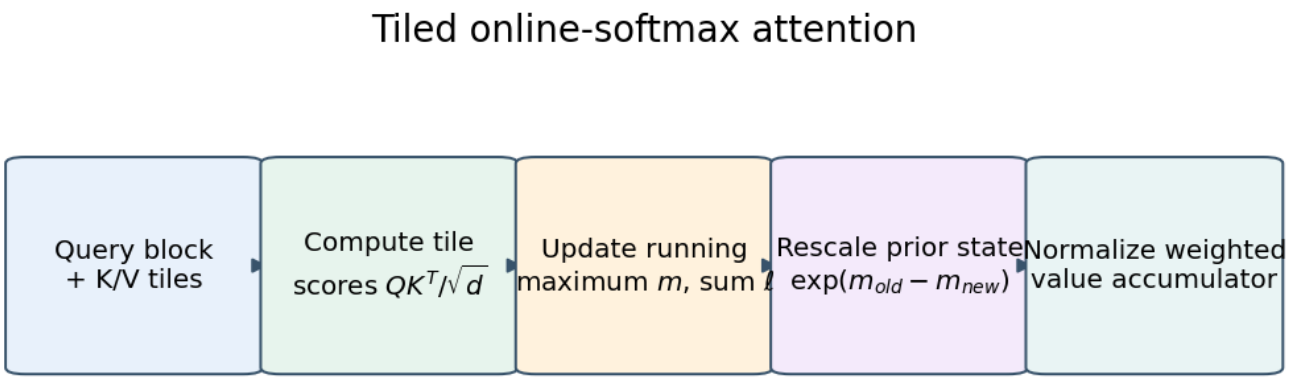


Fig. 4. Tiled attention maintains an online softmax state and rescales it when the running maximum changes.

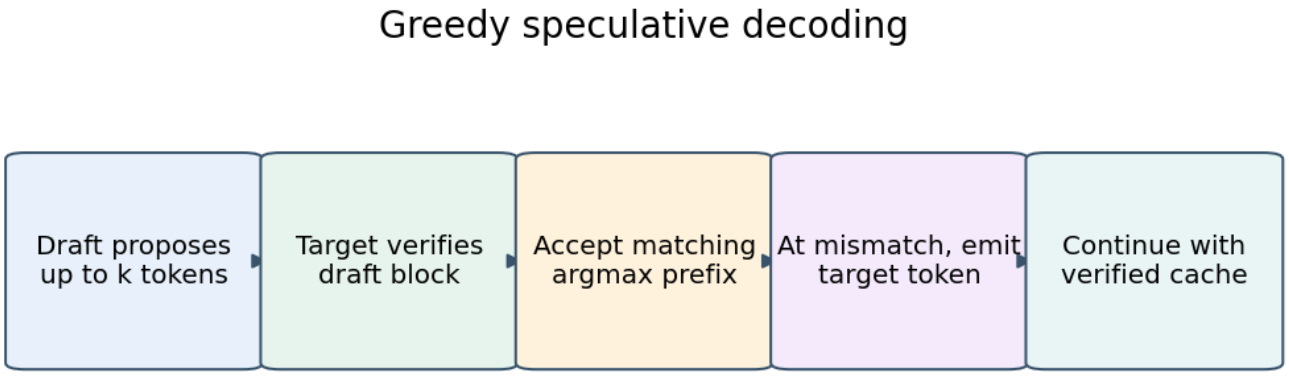


Fig. 5. Greedy speculative decoding proposes a block, accepts the matching prefix, and uses the target token at the first mismatch.

For the final test split, criterion-level accuracy and macro-F1 receive percentile bootstrap 95% intervals from 1,000 resamples with seed 42. Evidence extraction reports micro scores and primary macro scores over records with non-empty expert evidence; empty-gold abstention accuracy and invalid sentence IDs are reported separately.

KV-cache benchmarks use deterministic synthetic token IDs, two warm-ups, and five timed repeats per grid point; they compare full-prefix recomputation with cached decoding. Tiled attention uses deterministic random query/key/value tensors and records accuracy and performance grids, including unavailable fused backends and out-of-memory cases. Speculative-decoding benchmarks use deterministic synthetic prompts, two warm-ups, five repeats, two output lengths, and multiple values of k. These synthetic workloads measure inference mechanics, not clinical matching quality.

The final evaluation uses greedy no-cache decoding with 64 generated tokens on the four predeclared base/adapter combinations. The final kernel requires a verified freeze manifest for tag `v1.0-frozen`; each pair is guarded against repeat access in the access log.

VII. Results

A. Classification and evidence

Table III is generated directly from the four recovered validation outputs and the one-time final test outputs. The headline test values and their bootstrap intervals are reported in the abstract. Figure 6 compares accuracy and fixed-label macro-F1 with criterion-level bootstrap intervals; Fig. 7 shows per-class behavior. Confusion matrices are saved for every evaluated model; the headline model's test confusion matrix is included below.

TABLE III
CRITERION-LEVEL BOOTSTRAP 95% CONFIDENCE INTERVALS (1,000 RESAMPLES; SEED 42).

Split	Model	Accuracy	Macro-F1	Accuracy CI	Macro-F1 CI
TEST	Base 0.5B	0.325	0.093	[0.262, 0.393]	[0.071, 0.122]
TEST	Base 1.5B	0.204	0.158	[0.150, 0.262]	[0.126, 0.188]
TEST	Fine-tuned 0.5B	0.694	0.410	[0.636, 0.752]	[0.347, 0.469]
TEST	Fine-tuned 1.5B	0.762	0.476	[0.704, 0.816]	[0.420, 0.529]
VAL	Base 0.5B	0.257	0.084	[0.204, 0.320]	[0.061, 0.111]
VAL	Base 1.5B	0.267	0.204	[0.209, 0.330]	[0.165, 0.243]
VAL	Fine-tuned 0.5B	0.714	0.432	[0.650, 0.777]	[0.365, 0.487]
VAL	Fine-tuned 1.5B	0.748	0.550	[0.689, 0.806]	[0.394, 0.646]

TABLE IV
EVIDENCE-SET SCORES; PRIMARY MACRO-F1 AVERAGES RECORDS WITH NON-EMPTY EXPERT EVIDENCE.

Split	Model	Micro-F1	Primary macro-F1	Empty gold	Abstention acc.
TEST	Base 0.5B	0.020	0.007	145	1.000
TEST	Base 1.5B	0.242	0.498	145	0.000
TEST	Fine-tuned 0.5B	0.473	0.526	145	0.890
TEST	Fine-tuned 1.5B	0.542	0.584	145	0.897
VAL	Base 0.5B	0.030	0.009	116	0.991
VAL	Base 1.5B	0.216	0.409	116	0.000
VAL	Fine-tuned 0.5B	0.420	0.470	116	0.888
VAL	Fine-tuned 1.5B	0.546	0.616	116	0.853

Fine-tuning is the dominant factor. On the held-out test patients, LoRA fine-tuning raises accuracy from 0.204 to 0.762 for the 1.5B model and from 0.325 to 0.694 for the 0.5B model, and fixed-label macro-F1 from 0.158 to 0.476 and from 0.093 to 0.410. Both fine-tuned models produced parseable JSON for all 206 test criteria, while the base 0.5B model failed to parse on 1.0% of them. The base models fail in characteristic ways. Base 1.5B labels 109 of 206 test criteria *not applicable* (gold: 15) and predicts *excluded* 32 times when no test criterion carries that gold label. Base 0.5B almost always abstains. Model scale matters less than supervision: the fine-tuned 0.5B outperforms the base 1.5B by a wide margin, and the 1.5B adds about 7 accuracy points over the fine-tuned 0.5B. Validation and test results agree closely (fine-tuned 1.5B accuracy 0.748 vs. 0.762), so the selected checkpoint does not appear over-fitted to the validation patients. For evidence, the fine-tuned 1.5B reaches micro-F1 0.542 and primary macro-F1 0.584 (Table IV), and no model cited a sentence ID outside the note.

The validation comparison selected the final fine-tuned 1.5B adapter as the headline model (fixed-label macro-F1 0.5500); this is a comparison among final adapters, not per-epoch checkpoint selection. Evidence metrics and empty-gold conventions are shown in Table IV.

B. Training dynamics

Figure 9 plots the logged optimization loss and validation loss. The available histories do not contain generation-based F1 for each epoch, so no epoch-level F1 curve is inferred.

C. Inference benchmarks

The KV-cache results report latency, decode throughput, peak allocation, and measured cache growth; Table VI and Figs. 10–11 are generated from the benchmark JSON. The

TABLE V
PER-CLASS PRECISION, RECALL, AND F1 FOR TEST FT 1P5B.

Label	Precision	Recall	F1	Support
included	0.958	0.852	0.902	27
not included	0.000	0.000	0.000	7
not enough information	0.736	0.591	0.655	66
not applicable	0.714	0.333	0.455	15
excluded	0.000	0.000	0.000	0
not excluded	0.738	0.989	0.845	91

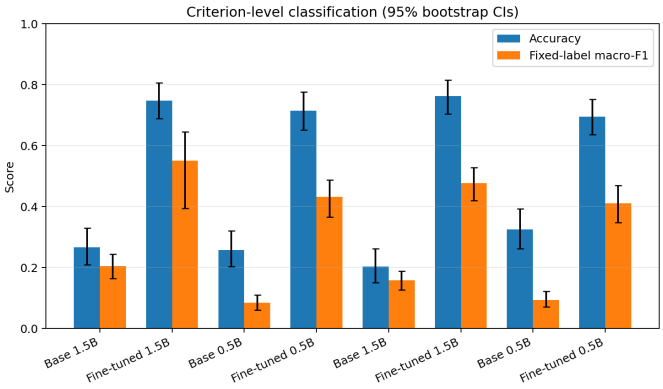


Fig. 6. Accuracy and fixed-label macro-F1 across validation and final test evaluations.

tiled-attention error table (Table VII) and performance plots distinguish successful, unavailable, and OOM rows. Speculative acceptance, speedup, and latency are plotted against k separately for each draft and output length (Table VIII). No speedup is asserted without the corresponding measured result.

All benchmarks ran on one Kaggle Tesla T4 (PyTorch 2.11, Transformers 4.57.6) with one warm-up and three timed repeats for KV and attention, and two repeats for speculative decoding. *KV cache.* For the fine-tuned 1.5B model in float16, generating 128 tokens after a 1024-token prompt took 46.4 s with full-prefix recomputation and 4.72 s with our KV cache (2.76 vs. 29.1 output tokens/s; 9.8×). The gain grows with context: 1.5×, 4.9× and 9.8× at 128, 512 and 1024 input tokens (1.1×, 2.5× and 6.2× for the 0.5B model). Measured cache memory grows linearly at 28 KiB per token for the 1.5B model (29.4 MB at 1024 tokens), consistent with 28 layers × 2 KV heads × 128 dimensions in float16, and peak allocation fell from 3.81 GB to 3.47 GB. All 24 precision checks per model produced identical greedy tokens with and without the cache. *Tiled attention.* All 80 accuracy configurations (5 shapes, causal and non-causal, float32 and float16, block sizes 16–128) were within tolerance; the maximum absolute error against the float32 reference was 1.2×10^{-6} in float32 and 9.7×10^{-4} in float16. Our tiled implementation is written in PyTorch rather than a fused kernel, so it is slower than the reference (0.33 s vs. 0.03 s at 2048 tokens, head dimension 128). Its value is memory: at 2048 tokens it peaked at 93 MB against 878 MB for the naive reference, and at 8192 tokens the reference ran out of memory while the tiled version completed

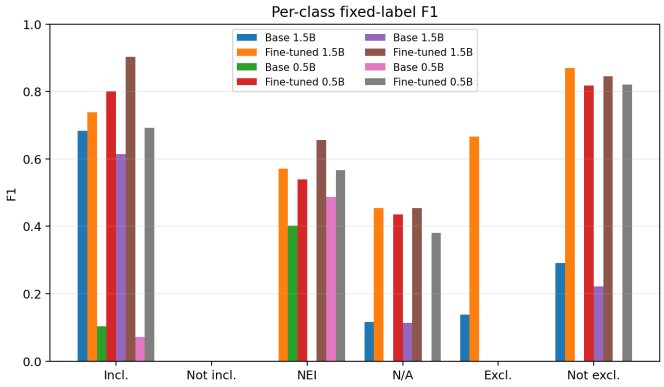


Fig. 7. Per-class F1 across the evaluated base and fine-tuned variants.

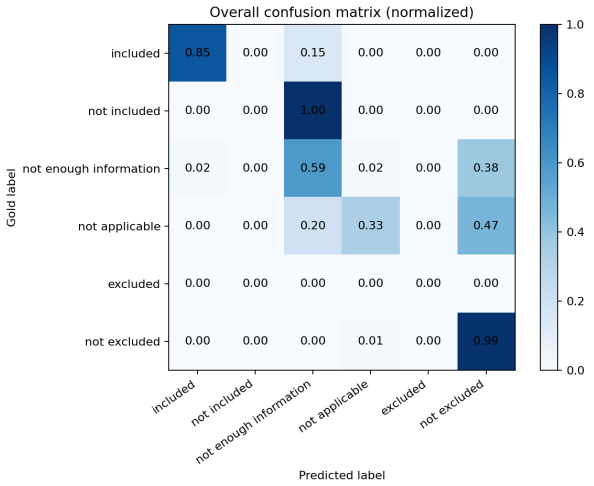


Fig. 8. Row-normalized six-label confusion matrix for the fine-tuned 1.5B headline model on the fixed test set.

using 345 MB. PyTorch's memory-efficient SDPA kernel remains far faster (0.020 s at 8192 tokens); FlashAttention is unavailable on the T4 (sm_75). *Speculative decoding.* Over 6 synthetic prompts and 64 new tokens, speculative output matched target greedy decoding exactly (token and sequence equality 1.0, zero mismatches) for every k. Acceptance fell from 0.80 (k=1) to 0.46 (k=8) with the base 0.5B draft and from 0.83 to 0.48 with the fine-tuned draft, so the in-domain draft is accepted slightly more often. Speculation cut target forward passes from one per token to 0.23–0.57 per token, yet wall-clock speedup was below 1 (0.94× at best, 0.60× at k=8). At batch size one on a T4, a 0.5B draft has about a third of the target's parameters but needs one sequential forward pass per proposed token, and our Python-level loop adds per-iteration overhead, so the saved target passes do not offset the draft passes. We report this negative result as measured; a much smaller draft or a fused verification kernel would be needed for a real speedup.

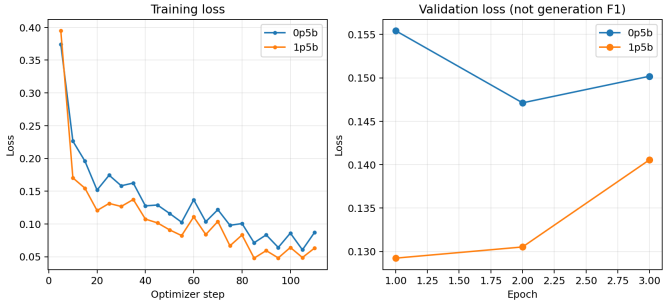


Fig. 9. Training and recorded validation loss; validation generation F1 was not logged per epoch.

TABLE VI
KV-CACHE LATENCY SPEEDUP ON DETERMINISTIC SYNTHETIC TOKEN IDS.

Run	Input tokens	Output tokens	Dtype	Speedup
kv_0p5b	128	128	float16	1.11
kv_0p5b	512	128	float16	2.49
kv_0p5b	1024	128	float16	6.24
kv_1p5b	128	128	float16	1.51
kv_1p5b	512	128	float16	4.85
kv_1p5b	1024	128	float16	9.82
kv_tiny	128	128	float32	1.08
kv_tiny	512	128	float32	1.08
kv_tiny	1024	128	float32	1.36

D. Error analysis

The automated error analyzer selects distinct candidate annotations from final predictions using label and evidence discrepancies. Its diagnoses are provisional, and note text is not copied into tracked report files. Table IX summarizes candidates for author review; the full review worksheet is `docs/ERROR_ANALYSIS.md`. The fine-tuned 1.5B model is wrong on 49 of 206 test criteria. Its errors are almost entirely *abstention-boundary* errors; it never swaps the two decisive labels of a criterion (e.g., *included* vs. *not included*). The largest group (25 errors, 51%) is *not enough information* → *not excluded* on exclusion criteria: the note does not mention a condition (e.g., a GFR threshold or previous axillary surgery), and the model treats absence as negation. This is the unsafe direction for screening, which is why decisive labels without cited evidence are routed to *needs review*. The second group (7 errors) is *not included* → *not enough information*. For example, for "patients undergoing elective carotid endarterectomy" the model cites the correct procedure sentence but does not conclude that a different procedure fails the criterion; it never predicts *not included* on the test set. Further cases involve qualitative versus numeric thresholds ("tachycardic" for heart rate $\geq$ 90), conditional criteria ("if female, using birth control"), and *not applicable* versus *not excluded* on sex-specific criteria, where expert labels themselves are inconsistent. In 27 correctly labelled rows the cited evidence differs from the expert set, so label accuracy overstates explanation quality.

TABLE VII
LARGEST OBSERVED ABSOLUTE ATTENTION ERRORS ACROSS THE CONFIGURED CORRECTNESS GRID.

Dtype	Q length	K length	Head dim	Block	Max abs. error	Within tol.
float16	512	512	128	16	0.000971	True
float16	512	512	128	32	0.000971	True
float16	512	512	128	64	0.000971	True
float16	512	512	128	128	0.000971	True
float16	129	129	64	16	0.000952	True
float16	129	129	64	32	0.000952	True
float16	129	129	64	64	0.000952	True
float16	129	129	64	128	0.000952	True
float16	64	64	32	16	0.000775	True
float16	64	64	32	32	0.000775	True
float16	64	64	32	64	0.000775	True
float16	64	64	32	128	0.000775	True

TABLE VIII
SPECULATIVE DECODING MEANS OVER SYNTHETIC PROMPTS, GROUPED BY OUTPUT LENGTH.

Draft	Output tokens	k	Acceptance	Speedup
base_0.5B	64	1	0.796	0.93
base_0.5B	64	2	0.730	0.88
base_0.5B	64	4	0.622	0.77
base_0.5B	64	8	0.461	0.60
finetuned_0.5B	64	1	0.834	0.94
finetuned_0.5B	64	2	0.747	0.89
finetuned_0.5B	64	4	0.628	0.78
finetuned_0.5B	64	8	0.482	0.62

VIII. APPLICATION, DEPLOYMENT, AND LIMITATIONS

The Gradio application presents criterion screening, evidence, an inference lab, benchmarks, and offline results. It is officially deployed as a locally hosted browser application, which satisfies the assignment. On Windows, `run_app.bat` starts the app; the first run downloads the selected public model. An optional temporary Gradio share link is intended only for defense-day remote access. The Hugging Face Space remains staged but was not deployed because the platform requires a PRO account for the selected 2026 hardware.

The dataset contains a limited number of patients and the held-out split is correspondingly small; bootstrap intervals are included to communicate uncertainty. Labels and evidence inherit the source annotation's limitations, while evidence disagreement may reflect incomplete annotation rather than a model error. Synthetic-token inference benchmarks do not predict clinical workload latency. The application uses deterministic templates rather than independently verified rationales, and its aggregation rule is explicitly preliminary. CriterionLens is not a clinical decision system, must not be used for enrollment decisions, and requires expert review of every case.

IX. CONCLUSION

CriterionLens demonstrates a reproducible, evidence-oriented pipeline for criterion-level trial screening with small instruction models, together with explicit inference implementations and a locally hosted demonstration. The results are limited to a single public annotation source and a small patient-disjoint evaluation. The KV cache gave large measured gains that grow with context, whereas exact speculative decoding did not pay off with a 0.5B draft on a T4, which shows why inference optimizations must be measured on the target hardware. Further work should test additional cohorts, improve annotation review, and optimize inference with measured rather than assumed benefits. Any clinical use would require substantially broader validation, governance, and oversight.

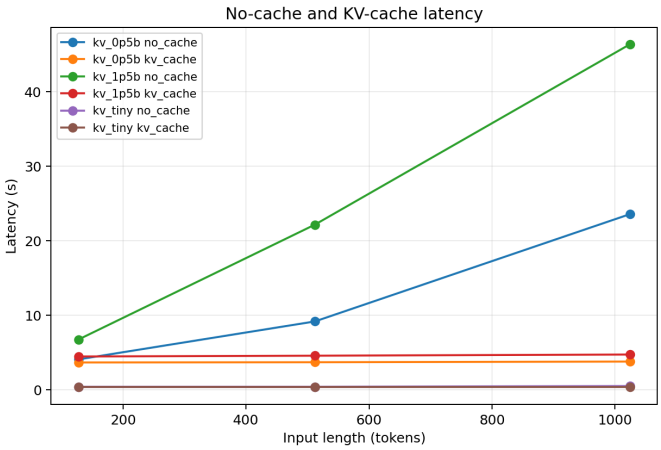


Fig. 10. Full-prefix and KV-cache latency on synthetic token IDs.

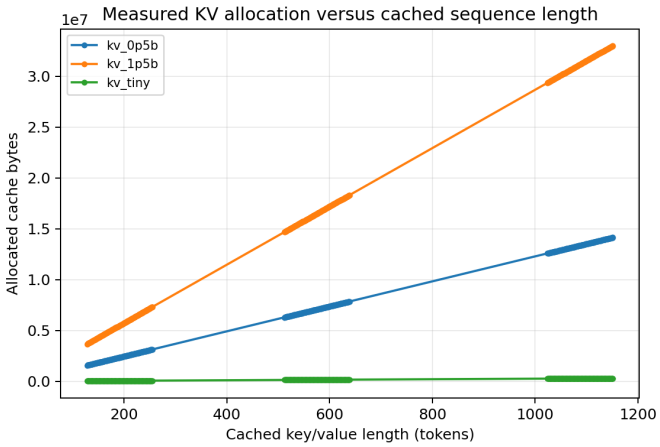


Fig. 11. Measured key/value allocation versus cached sequence length.

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TABLE IX
PROVISIONAL ERROR CANDIDATES FOR AUTHOR REVIEW; NOTE TEXT IS OMITTED.

ID	Variant/type	Gold → predicted	Candidate analysis
169	ft_1p5b/exclusion	not enough information → not excluded	missing_information_guess, unsupported_decision: The expert label is not-enough-information while the model made a decisive judgment; inspect whether the note actually supports that judgment. The output marks a decisive judgment without cited evidence; this should remain a manual-review case.
174	ft_1p5b/exclusion	not enough information → not excluded	missing_information_guess, unsupported_decision: The expert label is not-enough-information while the model made a decisive judgment; inspect whether the note actually supports that judgment. The output marks a decisive judgment without cited evidence; this should remain a manual-review case.
179	ft_0p5b/exclusion	not excluded → not excluded	unsupported_decision, missed_gold_evidence: The output marks a decisive judgment without cited evidence; this should remain a manual-review case. The cited evidence does not overlap the expert evidence; inspect the sentence numbering and whether the model missed the supporting fact.
166	ft_1p5b/exclusion	not excluded → not excluded	evidence_set_discrepancy: The model cited sentence IDs outside the expert evidence set; assess relevance locally because expert evidence may not be exhaustive.
193	ft_1p5b/exclusion	not enough information → not excluded	missing_information_guess, unsupported_decision, missed_gold_evidence: The expert label is not-enough-information while the model made a decisive judgment; inspect whether the note actually supports that judgment. The output marks a decisive judgment without cited evidence; this should remain a manual-review case. The cited evidence does not overlap the expert evidence; inspect the sentence numbering and whether the model missed the supporting fact.
307	ft_1p5b/inclusion	not enough information → not enough information	evidence_set_discrepancy: The model cited sentence IDs outside the expert evidence set; assess relevance locally because expert evidence may not be exhaustive.
190	ft_0p5b/inclusion	not enough information → not enough information	evidence_set_discrepancy: The model cited sentence IDs outside the expert evidence set; assess relevance locally because expert evidence may not be exhaustive.
305	ft_1p5b/inclusion	not enough information → not enough information	evidence_set_discrepancy: The model cited sentence IDs outside the expert evidence set; assess relevance locally because expert evidence may not be exhaustive.
315	base_1p5b/exclusion	not enough information → excluded	missing_information_guess, evidence_set_discrepancy: The expert label is not-enough-information while the model made a decisive judgment; inspect whether the note actually supports that judgment. The model cited sentence IDs outside the expert evidence set; assess relevance locally because expert evidence may not be exhaustive.
323	ft_0p5b/inclusion	not enough information → included	missing_information_guess: The expert label is not-enough-information while the model made a decisive judgment; inspect whether the note actually supports that judgment.
325	ft_1p5b/inclusion	not enough information → not enough information	evidence_set_discrepancy: The model cited sentence IDs outside the expert evidence set; assess relevance locally because expert evidence may not be exhaustive.
389	ft_1p5b/exclusion	not enough information → not excluded	missing_information_guess, unsupported_decision: The expert label is not-enough-information while the model made a decisive judgment; inspect whether the note actually supports that judgment. The output marks a decisive judgment without cited evidence; this should remain a manual-review case.

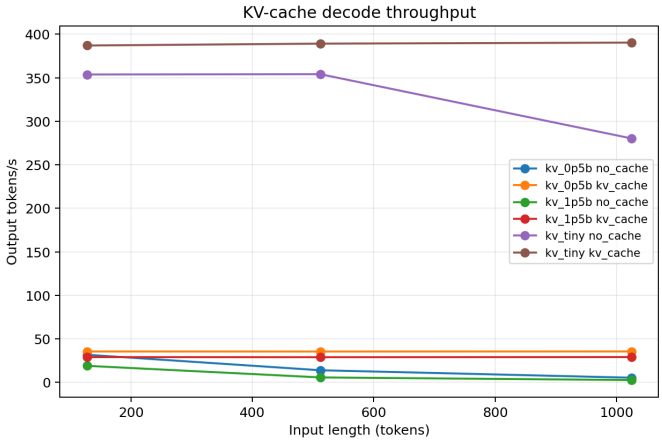


Fig. 12. KV-cache decode throughput on synthetic token IDs.

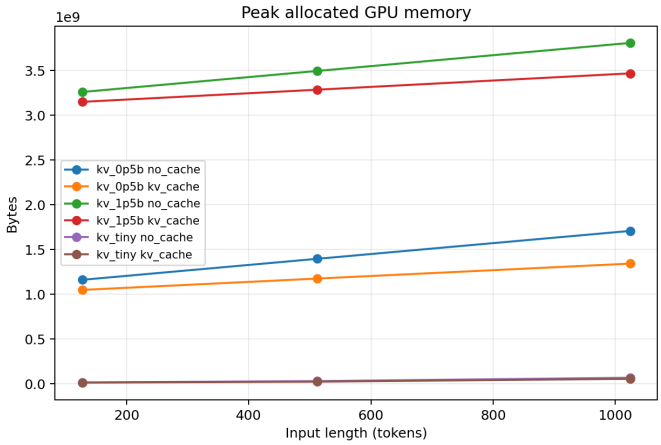


Fig. 13. Peak memory allocation for no-cache and KV-cache decoding.

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[8] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "Qlora: Efficient finetuning of quantized llms," *arXiv preprint arXiv:2305.14314*, 2023.

[9] M. Milakov and N. Gimelshein, "Online normalizer calculation for softmax," *arXiv preprint arXiv:1805.02867*, 2018.

[10] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré, "Flashattention: Fast and memory-efficient exact attention with io-awareness," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[11] W. Kwon *et al.*, "Efficient memory management for large language model serving with pagedattention," in *Proceedings of the 29th Symposium on Operating Systems Principles*, 2023.

[12] Y. Leviathan, M. Kalman, and Y. Matias, "Fast inference from transform-

Largest tiled-attention numerical errors

Dtype	Q/K	Head dim	Block	Max abs err.	Within tol.
float16	512/512	128	16	0.00097	True
float16	512/512	128	32	0.00097	True
float16	512/512	128	64	0.00097	True
float16	512/512	128	128	0.00097	True
float16	129/129	64	16	0.00095	True
float16	129/129	64	32	0.00095	True
float16	129/129	64	64	0.00095	True
float16	129/129	64	128	0.00095	True
float16	64/64	32	16	0.00077	True
float16	64/64	32	32	0.00077	True
float16	64/64	32	64	0.00077	True
float16	64/64	32	128	0.00077	True

Fig. 14. Worst observed numerical errors for tiled attention.

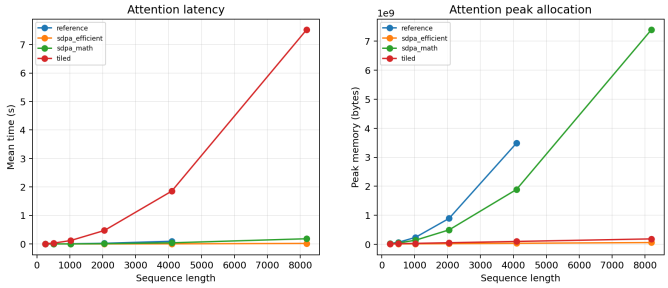


Fig. 15. Tiled, reference, and available fused attention performance on synthetic tensors.

ers via speculative decoding," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202, 2023, pp. 19 274–19 286.

Appendix A
Reproducibility

The repository records the dataset revision, patient split manifest, training configurations, model identifiers, benchmark configurations, evaluation metrics, and the final-test access log. Fast tests run with `.venv\Scripts\python.exe -m pytest -q`. Result tables and figures are regenerated with `.venv\Scripts\python.exe scripts\make_figures.py -require-test -require-bench`. Training archives and model weights are excluded from Git; published adapters and merged models are linked from the project README.

Artifacts. Code and execution instructions: https://github.com/SyedAshherMajid/criterion-lens (Windows start-up: `run_app.bat`). LoRA adapter (headline): https://huggingface.co/Ashher123/criterion-lens-qwen2.5-1.5b-lora; merged models: https://huggingface.co/Ashher123/criterion-lens-qwen2.5-1.5b-merged and https://huggingface.co/Ashher123/criterion-lens-qwen2.5-0.5b-merged. Split manifest: `data/splits/split_manifest.json`. Benchmark logs: `results/bench/`. Demonstration video: https://youtu.be/Z3mVr0ccfKE.

Appendix B
AI-use statement

Claude assisted with requirements analysis and preparation of the design and execution plan. An AI coding assistant using

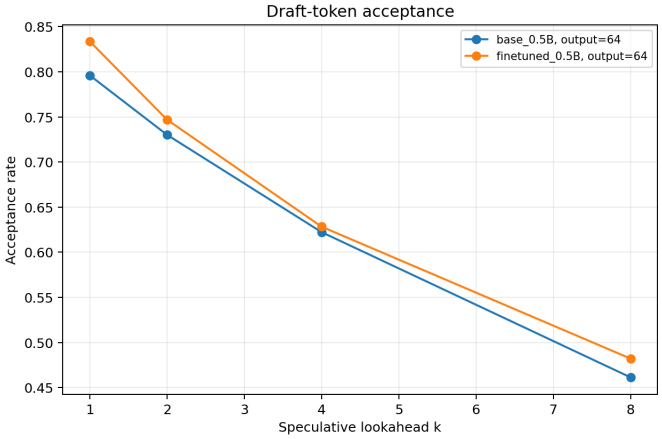


Fig. 16. Speculative draft-token acceptance by lookahead, draft model, and output length.

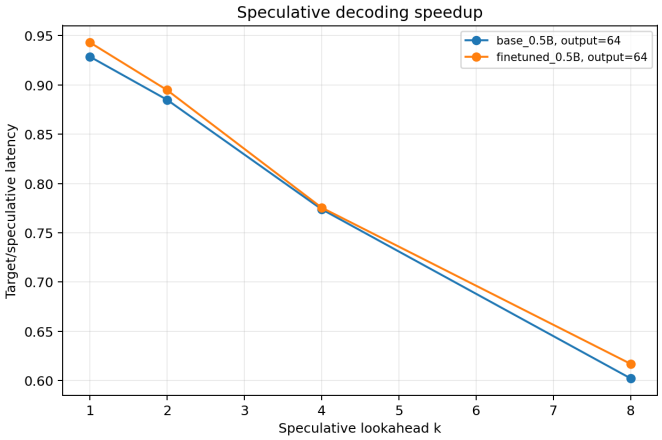


Fig. 17. Measured target-to-speculative latency ratio; values below one indicate slowdown.

Copilot SDK in VS Code assisted with implementation, tests, and documentation. Claude Code assisted in the final phase with running the frozen test evaluation and benchmarks on Kaggle, the error-analysis write-up, the results text of this paper, and the scripted, captioned screen recording of the demonstration. Experimental values in this paper are generated from the repository's recorded result files; citations are taken from the verified literature notes. The author remains responsible for reviewing the paper, the analysis, and every claim before submission.

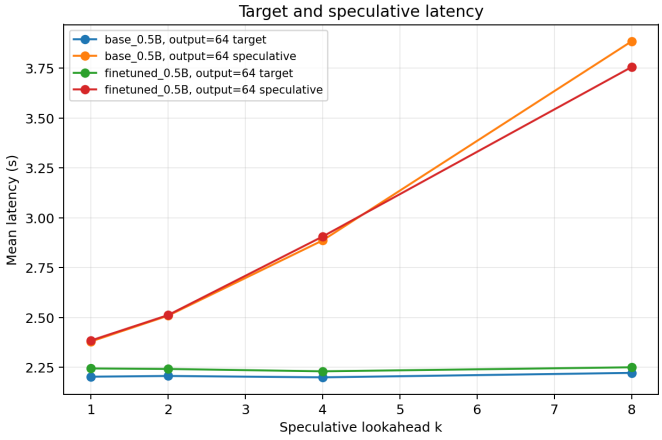


Fig. 18. Target and speculative latency by lookahead.